Phishing Email Detection & Awareness Report

Prepared by: **Elvina Prakash Pillay**

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1. Introduction

Phishing is one of the most common and damaging cyber threats faced by individuals and organizations today. Attackers use deceptive emails to trick recipients into clicking malicious links, downloading harmful files, or revealing sensitive information such as passwords and payment details. These attacks often succeed not because of weak systems, but because users are unaware of how to recognize suspicious emails.

This report's objectives are to examine a collection of phishing email samples, highlight typical phishing warning signs, and offer precise advice to assist users in avoiding becoming targets of such attacks. Organizations can utilize this report's professional structure to illustrate email risk classification, technical header analysis, and awareness tactics for security training and teaching.

2. Tools Used

The following tools were utilized to analyze phishing emails and document findings:

i. **Google Admin Toolbox - Messageheader Analyzer**
Used to examine email headers.

ii. **MXToolbox Email Header Analyzer**
Verified email routing, authentication, and domain reputation.

iii. **Web Browser & Safe Link Inspection**
Checked suspicious URLs without clicking on them, using hover-over inspections and safe online link-check tools.

iv. **Microsoft Word**
Documented findings and created a professional report for awareness purposes.

3. Executive Summary: Phishing Email Samples

The following table provides a high-level overview of the phishing emails analyzed in this report, summarizing key indicators and risk levels for quick reference. This section allows readers to quickly understand the scope of the analyzed threats before reviewing detailed sample analyses in Section 4.

Note: *URLs in this report use [.] instead of . to prevent accidental clicks.*

#	Email Subject	Sender Email	Key Phishing Indicators	Risk Classification
1	Action Required: PayPal Payment Request	support@paypal-support-alert.com	Suspicious domain; urgent language; suspicious link; generic greeting; reply-to mismatch	**Phishing**

2	Your Package is Waiting for Delivery	tracking@express-shipping-alert.com	Suspicious domain; urgency; suspicious link; generic greeting	**Phishing**
3	You received a secure document	noreply@secure-share-docs.com	Suspicious domain; urgency; suspicious login link; generic greeting	**Phishing**
4	Netflix Account Suspended — Update Payment	billing@netflix-support-center.com	Suspicious domain; urgency/fear language; suspicious link; generic greeting; reply-to mismatch	**Phishing**
5	DHL Shipment Notification	notification@dhl-delivery-alert.net	Suspicious domain; urgency; suspicious download link; generic greeting; attachment/download request	**Phishing**

Section 4 provides a detailed examination of each phishing email, including indicators, header checks, and risk classification.

4. Email Sample Analysis

4.1 Sample 1 - PayPal Payment Request Phishing Email

Subject: Action Required: PayPal Payment Request
Sender Email: support@paypal-support-alert.com

Summary:
This email informs the recipient that a payment request has been made on their PayPal account and asks them to use a link to confirm or reject it immediately. In an effort to trick the user into clicking the link, it attempts to cause panic and a sense of urgency.

Key Phishing Indicators

- **Suspicious sender domain:** paypal-support-alert.com is not the official PayPal domain.
- **Urgent, fear-based language:** "Failure to respond within 24 hours may result in automatic processing of the payment."
- **Suspicious link:** http://paypal-secure-resolution[.]com/verify — does not belong to PayPal.
- **Generic greeting:** "Dear Customer" instead of the user's real name.
- **Reply-to mismatch:** Points to attacker@malicious-server.ru, which is unrelated to PayPal.

Header Analysis

Key Findings:

- SPF authentication failed, indicating the sending IP address is not authorized to send emails for the domain.
- No DKIM signature was found, meaning the email lacks a cryptographic signature verifying the sender's identity.
- DMARC authentication failed or no DMARC policy was found for the domain.
- The sending IP address was flagged on a blacklist, suggesting the server may have been previously associated with malicious activity.

Conclusion:

The authentication failures and the absence of proper email security configurations indicate that the email was not sent from a legitimate server. These technical indicators strongly suggest that the message is a phishing email designed to impersonate a trusted service.

Risk Classification
Phishing

4.2 Sample 2 - Package Delivery Phishing Email

Subject: Your Package is Waiting for Delivery
Sender Email: tracking@express-shipping-alert.com

Summary:
This email requests that the receiver confirm their address via a tracking portal link after claiming that a package has reached a distribution location. It puts the user under pressure to act quickly by creating a sense of urgency, which is a common phishing approach.

Key Phishing Indicators

- **Suspicious sender domain:** express-shipping-alert.com is not affiliated with any official courier service.
- **Urgency:** The email states the address must be confirmed within 12 hours, pressuring immediate action.
- **Suspicious link:** http://track-delivery-service[.]xyz/confirm — the domain is unfamiliar and unrelated to real shipping companies.
- **Generic greeting:** “Dear Customer” instead of the recipient’s actual name.

Risk Classification
Phishing

4.3 Sample 3 - Secure Document Sharing Phishing Email

Subject: You received a secure document
Sender Email: noreply@secure-share-docs.com

Summary:
This email requests that the user log in using a link to access a secure document that was sent to them via OneDrive. It creates a sense of urgency ("link will expire today") and presents itself as official to fool the user into disclosing login information.

Key Phishing Indicators

- **Suspicious sender domain:** secure-share-docs.com is not affiliated with Microsoft or OneDrive.
- **Urgency:** The message warns that the link will expire today, pressuring the user to act quickly.
- **Suspicious login link:** http://onedrive-secure-docs[.]site/login — redirects to an unknown domain.
- **Generic greeting:** "Hello" instead of addressing the recipient by name.

Risk Classification
Phishing

4.4 Sample 4 - Netflix Billing Phishing Email

Subject: Netflix Account Suspended — Update Payment
Sender Email: billing@netflix-support-center.com

Summary:
According to this email, a failed payment has resulted in the suspension of the recipient's Netflix account. In order to trick the user into clicking, it creates a sense of urgency and panic by urging them to change their payment details right away via a link.

Key Phishing Indicators

- **Suspicious sender domain:** netflix-support-center.com is not the official Netflix domain.
- **Urgency & fear-based language:** "Your account may be permanently closed if payment details are not updated within 24 hours."
- **Suspicious link:** http://netflix-billing-update[.]live/account — the domain is unrelated to Netflix.
- **Generic greeting:** "Dear Netflix User" instead of addressing the recipient by name.
- **Reply-to mismatch:** payments@account-update.ru — points to an unrelated domain.

Header Analysis

Key Findings:

- **SPF authentication softfail:** The sending IP address (103.145.67.82) is not authorized to send emails for netflix-support-center.com.

- **No DKIM signature:** The email lacks a cryptographic signature to verify the sender's identity.
- **DMARC authentication failed:** No valid DMARC policy found for the domain, meaning domain-based authentication is missing.
- **Blacklist status:** The sending IP is listed on a blacklist, indicating possible prior malicious activity.

Conclusion:
The email was not sent from a legitimate Netflix server. SPF and DMARC failures, the missing DKIM signature, and the blacklisted sending IP confirm that the email is a phishing attempt designed to impersonate Netflix.

Risk Classification
Phishing

4.5 Sample 5 - DHL Shipment Notification Phishing Email

Subject: DHL Shipment Notification
Sender Email: notification@dhl-delivery-alert.net

Summary:
This email requests that the receiver download the shipping paperwork in order to reschedule delivery, claiming that DHL tried delivery. As a frequent phishing tactic, it instills a sense of urgency and convinces the user to click a link or download a file.

Key Phishing Indicators

- **Suspicious sender domain:** dhl-delivery-alert.net is not an official DHL domain.
- **Urgency & fear-based language:** Suggests immediate action to avoid losing the package.
- **Suspicious download link:** http://dhl-shipment-docs[.]info/download — not affiliated with DHL.
- **Generic greeting:** "Dear Customer" instead of the recipient's actual name.
- **Attachment/download request:** Phishing emails often include fake documents to steal information or deliver malware.

Risk Classification
Phishing

5. Prevention & Awareness Guidelines

To protect against phishing attacks, users should follow these best practices:

i. Verify the Sender

- Always check the sender's email address carefully.

- Be cautious if the domain is unfamiliar or slightly misspelled.

ii. Inspect Links Before Clicking

- Hover over links to see the real URL before clicking.
- Avoid clicking on links that look suspicious or point to unknown domains.
- Never download attachments from untrusted sources.

iii. Look for Red Flags

- Be alert to urgent or fear-based language that pressures immediate action.
- Generic greetings like "Dear Customer" are common in phishing emails.
- Requests for sensitive information (passwords, payment info, OTPs) via email are almost always phishing attempts.

iv. Use Email Authentication Features

- Check for SPF, DKIM, and DMARC authentication results when possible.
- Emails failing these checks should be treated as suspicious.

v. Keep Software and Security Updated

- Keep operating systems, browsers, and email clients up to date to reduce vulnerabilities.
- Use antivirus and anti-malware tools to help detect malicious content.

vi. Report Suspicious Emails

- Forward suspected phishing emails to your IT or security team.
- For personal accounts, report phishing emails to the company being impersonated (e.g., PayPal, Netflix, DHL).

vii. Educate and Train Users

- Conduct regular cybersecurity awareness sessions.
- Teach employees to recognize phishing tactics, verify URLs, and question unexpected emails.

6. Conclusion

Phishing remains one of the most significant cybersecurity threats because it targets human behaviour rather than technology. This report demonstrates how technical analysis, combined with awareness of common phishing tactics, can effectively identify and mitigate risks. By applying the prevention guidelines, users and organizations can significantly reduce the likelihood of falling victim to phishing attacks. Continuous education, vigilance, and proactive reporting are essential to maintaining email security and safeguarding sensitive information.

7. Glossary of Terms

- **Phishing:** A type of cyberattack where attackers trick users into giving sensitive information via fake emails or websites.

- **SPF (Sender Policy Framework):** A protocol that verifies if an email is sent from an authorized server for the sender's domain.

- **DKIM (DomainKeys Identified Mail):** A digital signature that confirms the email was not altered during transit.

- **DMARC (Domain-based Message Authentication, Reporting & Conformance):** A policy that tells email servers how to handle unauthenticated emails.

- **Suspicious Link:** A URL that may redirect to a malicious or fake website.

- **Generic Greeting:** Non-specific greetings like “Dear Customer” instead of addressing the recipient by name.

- **Urgent Language:** Phrases designed to pressure the recipient into immediate action, often used in phishing attacks.

- **Header Analysis:** The process of examining the technical routing and authentication information in an email header to verify legitimacy.

- **Attachment/Download Request:** An email request to open or download a file. Phishing emails often include fake documents to steal information or install malware.

- **Reply-to Mismatch:** When the email’s “Reply-to” address is different from the sender’s address, which may indicate the email is sent by an attacker.

- **Fake Domain / Lookalike Domain:** A web address that mimics a legitimate company to trick users into clicking or providing sensitive information.